What makes a HAR a HAR? Interpretable ML on human-accelerated noncoding regions

Martin Orkuma · MCB 540 – Functional Genomics, Spring 2026 · Central question: Which genomic features distinguish HARs from matched conserved non-accelerated elements?

1. Background & question

Human Accelerated Regions (HARs) are conserved noncoding sequences that changed rapidly on the human lineage after divergence from chimpanzees (Keough et al., 2023). Many have been linked to neurodevelopment and cognition.(Doan et al., 2016; Levchenko et al., 2018).

Hypothesis: HARs occupy distinctive regulatory contexts relative to matched conserved non-accelerated elements.

2. Approach

HAR set (Doan 2016) → Matched CNEs → 7 features → LR + RF → SHAP ranking

Dataset: 2,734 HARs vs 7,473 matched CNEs

Controls: length ±25%, mean phastCons ±0.05, ±10 kb buffer, coding exons excluded

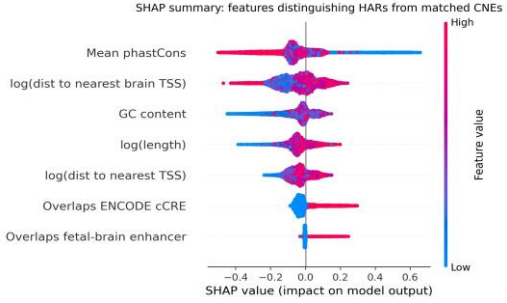
Models: logistic regression baseline + random forest, 5-fold stratified CV

Category	Features
Sequence	GC content, length
Conservation	Mean phastCons
Gene context	Distance to nearest TSS, distance to nearest brain-expressed TSS.
Regulatory	ENCODE cCRE overlap, fetal-brain enhancer overlap

References

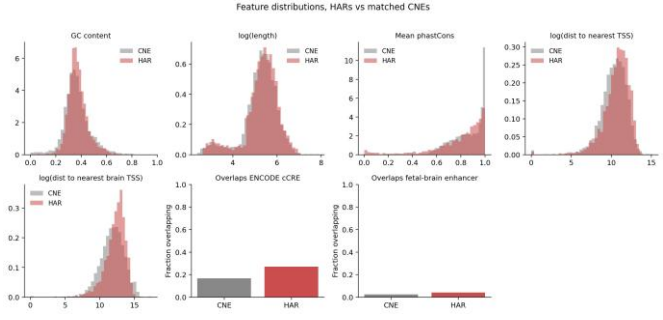
- Boyd et al. 2015
- Capra et al. 2013
- Cui et al. 2025
- Doan et al. 2016
- Keough et al. 2023
- Pollard et al. 2006

3. Result 1: HARs are distinguished by conservation, GC content, and regulatory overlap

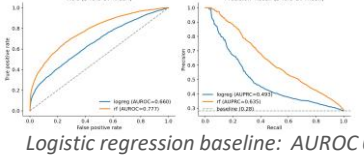


Top features: (1) lower phastCons → HAR; (2) ***farther*** from brain-expressed TSS → HAR; (3) higher GC; (4) cCRE overlap → HAR (HAR 27% vs CNE 17%).

4. Per-feature distributions (HAR vs matched CNE)



5. Result 3: Features moderately distinguish HARs from matched CNEs



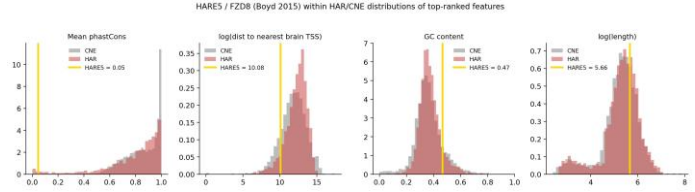
Random Forest

AUROC **0.777** ± 0.009

AUPRC **0.635** *(baseline 0.27)*

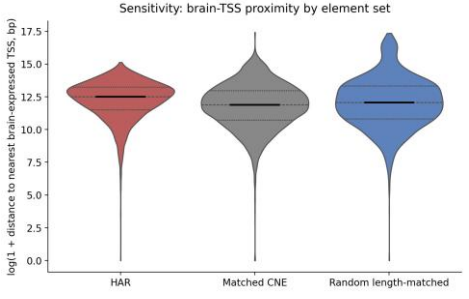
Logistic regression baseline: AUROC 0.660 · AUPRC 0.493 (blue line in ROC).

6. Case example: HARE5 / FZD8 shows acceleration



HARE5 / 2xHAR.238 (Boyd 2015; chr10:35,222,797–35,223,084, hg38) — phastCons **0.045**, an extreme outlier consistent with strong human-lineage acceleration.

7. Result 2: HARs are farther from brain-expressed TSSes



Median dist to brain TSS:

HAR: 269 kb

Random: 173 kb

CNE: 145 kb

8. Conclusions

- **HARs are regulatory-enriched:** HARs overlap cCREs more often than matched CNEs: 27% vs 17%.
- **HARs are not simply closer to brain-expressed genes:** Matched and random-control comparisons show HARs are farther from brain TSSes.
- **Conservation dominates the signal: Lower phastCons is the strongest distinguishing feature, but this is partly definitional.**
- **Caveat: Linear TSS distance does not prove regulatory targeting.**